

The *ReNo* Question Bank Methodology

*How the 94-item instrument is constructed, keyed, and scored
— and the principles every item obeys.*

Instrument: ReNo v1.1 — five-axis, 94 normative items

Form: Adult self-report (EN / RU) · parent-observer form forthcoming

Companion to: [The ReNo Methodology v1.1](#)

Version: 1.0 · Date: July 2026

STATUS: PILOT-READY, NOT YET VALIDATED

00 Scope

This document explains the *question bank* specifically: item format, axis mapping, the scoring key, and the eight construction rules that separate this bank from the legacy forced-choice test it replaces. It assumes the constructs defined in the main methodology whitepaper.

STATUS

The 94 items are a **pilot-ready pool**, not a finalized scale. The development study (item & factor analysis) is expected to drop the weakest items on each axis — that is the method working as designed, not failing. No score produced by this bank is validated until that study is complete.

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01 What changed from the legacy bank

The previous instrument was 94 forced-choice word-select items on four axes. It converts well but fights the methodology's scientific goals. Three structural changes define the new bank.

Dimension	Legacy bank	ReNo v1.1 bank
Scoring model	Forced-choice (ipsative) — pick one pole-word per item	Normative Likert — agree/disagree per statement. Enables factor analysis, Big-Five validation, and clean test-retest.
Item content	Single trait-words (“sociable” vs “reserved”)	Behaviourally anchored statements — observable actions, no trait labels.
Axes	4 (no Emotional Response)	5 — adds Emotional Response (18 items).
Item weights	Hand-assigned 2 / 1 / 0 per option	All items equal at pilot; discrimination established empirically, not assumed.
Languages	EN / RU / ES / FR / AR	EN / RU for pilot; localize the survivors post-analysis.

THE ONE HONEST TRADE-OFF

Forced-choice word-select is faster and lower-friction (Fogg's ability axis) — good for conversion. Normative Likert is validatable — good for science. This bank chooses science; if conversion friction proves costly, the fallback is to keep a forced-choice consumer form and run a normative form for the validation study only.

02 Item format & response scale

Every item is a first-person behavioural statement rated on one shared five-point agreement scale.

Value	English	Russian
1	Completely disagree	Совершенно не согласен(на)
2	Rather disagree	Скорее не согласен(на)
3	Hard to say	Трудно сказать
4	Rather agree	Скорее согласен(на)
5	Completely agree	Полностью согласен(на)

Each item carries three fixed attributes in the bank file: its **axis**, the **pole** that agreement scores toward, and a **reverse** flag. A worked example:

qid	axis	pole	reverse	item
Q001	EO	O	0	"After a party or large gathering, I leave with more energy than when I arrived."
Q011	EO	W	1	"After a few hours in company, I need time alone before I can do anything else."

Both items measure Energy Orientation; Q001 scores toward the Outward pole directly, Q011 is reverse-keyed (agreement indicates Inward). Mixing forward and reverse items within each axis is deliberate — see Rule 5.

03 The eight construction rules

Every item was written to obey these. They are the quality contract for the bank — any new item must pass all eight.

1. **Normative, not ipsative.** Independent agreement ratings, so scores can be correlated, factored, and validated.
2. **One construct per item.** No item may touch two axes. This is what keeps the factor structure clean and the stability read trustworthy.
3. **Behaviourally anchored.** Describe an observable action (“I finish tasks well before the deadline”), never a trait word (“I am organized”). People report behaviour accurately but infer traits poorly.
4. **Concrete and situational.** Each item names a recognisable moment — a party, a deadline, a disagreement — so the respondent recalls rather than self-theorises.
5. **Balanced keying.** Roughly half of each axis is reverse-keyed, so a respondent who agrees with everything scores near the midpoint. This neutralises acquiescence bias — the normative equivalent of what forced-choice got for free.
6. **Value-neutral poles.** Neither pole is written as better. No item frames a trait as a deficit (the legacy key's “too much detail, misses the point” is exactly what to avoid).
7. **Single, unambiguous idea.** No double-barrelled items (“I plan ahead *and* stay calm” secretly measures two axes). One verb, one behaviour.
8. **Contamination-safe by axis.** Composure and rumination appear only on Emotional Response; novelty-seeking only on Information Focus; no risk-propensity items anywhere; Structure Preference stays pure order-and-closure.

04 Axis map & item allocation

94 items across five axes, each axis pole-balanced. Reference pole = the “plus” direction a raw score of 100 points to.

Code	Axis	Poles (plus / minus)	Items	Split (plus / minus)
EO	Energy Orientation	O Outward / W Inward	19	10 O · 9 W
IF	Information Focus	C Concrete / A Abstract	19	10 C · 9 A
DB	Decision Basis	L Logic / V Values	19	10 L · 9 V
SP	Structure Preference	D Ordered / F Flexible	19	10 D · 9 F
ER	Emotional Response	S Steady / R Reactive	18	9 S · 9 R

Item ID blocks: EO Q001–Q019, IF Q020–Q038, DB Q039–Q057, SP Q058–Q076, ER Q077–Q094. Reference poles: **EO→O** **IF→C** **DB→L** **SP→D** **ER→S** .

05 The scoring key — item to axis to code

Four deterministic steps convert raw responses into the five-slot code. Everything runs on the continuous score; the code is display only.

Step 1 — item value

For an item with response r (1-5), normalise to 0-1 in the direction of the axis's plus pole:

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if reverse = 0: value = (r - 1) / 4
if reverse = 1: value = (5 - r) / 4
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Step 2 — axis raw score (0-100)

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X = mean(item values on that axis) × 100
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$X = 100$ means fully the plus pole, 0 fully the minus pole, 50 balanced.

Step 3 — pole, intensity, band

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pole      = plus pole if  $X \geq 50$ , else minus pole
intensity =  $|X - 50| \times 2$  (0-100, maps to the band rubric)
band digit = 1-20→1 · 21-40→2 · 41-60→3 · 61-80→4 · 81-100→5
```

Step 4 — assemble the code

Concatenate **[pole][band]** in fixed order EO·IF·DB·SP·ER, e.g. **W2 · A4 · V3 · F4 · S2**.

BALANCED BAND (DIGIT 0)

Band 0 is assigned when the score's **confidence interval spans 50** — i.e. the lean is within measurement error. Until reliability is established in the pilot, use a provisional rule of intensity ≤ 5 . This prevents labelling a near-balanced person with a false pole.

STORE THE SCORE, NOT THE BAND

Persist X (continuous) and every raw item response with a timestamp. All reliability, stability, and validation analysis runs on X . The band and code are regenerated from X for display — never the reverse.

06 Contamination control

The legacy instrument let three constructs leak across axes. Rule 8 enforces where each now lives, at the item level.

Content	Legacy location	ReNo v1.1 — enforced home
Composure / rumination ("невозмутим", replaying conflict)	Decision Basis (T/F)	→ Emotional Response only (Q077-Q094). DB items carry zero composure content.
Emotional stability / mood swings	Structure Preference (J/P)	→ Emotional Response only . SP is pure order-and-closure.
Novelty-seeking (conservative / innovator)	Structure Preference (J/P)	→ Information Focus (e.g. Q036 — "untested ways attract me"), where it is the same Openness signal.
Risk propensity	Structure Preference (J/P)	→ Dropped . A blend of three traits; measured nowhere, allowed to emerge in interpretation.

A concrete check: Decision Basis (Q039-Q057) contains only decision-criterion behaviour — weighing pros/cons vs weighing people, critiquing vs softening, rules vs exceptions. Nothing about staying calm. That content is entirely in Emotional Response, which is why the two axes can factor independently.

07 Known watch-items & the pilot plan

Honesty about the bank's soft spots, and the study that resolves them.

Watch-items (first drop candidates in analysis)

- **Q054** (“I sense the emotional temperature of the room”) — keyed to Decision Basis/Values, but may cross-load onto Emotional Response. If it does, drop it.
- **Q093** (“moved to tears by a film or music”) — deliberately captures the *positive* face of reactivity so Emotional Response is not an all-negative scale; verify it loads with the other R items and does not read as sadness alone.
- **Q073 / Q065** (workspace tidiness) — watch for overlap with cultural habit rather than trait Structure Preference.

The pilot plan (DeVellis sequence)

1. Administer all 94 to a development sample (the launch cohort is a viable first source).
2. Item analysis — drop low-discrimination and cross-loading items.
3. Confirmatory factor analysis — verify five clean factors.
4. Establish test-retest reliability (short interval) before any longitudinal claim.
5. Optimise to ~12 items per axis; localise the survivors to ES/FR/AR.
6. Convergent validation against NEO-PI-R / TIPI (traits), RIASEC (interests), Basic Psychological Needs Scale (motivation).

NEXT FORM TO BUILD

This is the **adult self-report** bank. The PsyID Youth product needs a parallel **parent-observer** bank — same constructs and key structure, child-world behaviours (school, play, homework), “My child...” phrasing, for ages 8–14.